Contraction Theory for Nonlinear Control

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Outline

What is Contraction Theory?

Contraction Theory is a powerful tool for analyzing the stability of nonlinear systems by examining differential dynamics, providing insights into the behavior of trajectories (not points) and their convergence properties [1].

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t) \quad \rightarrow \quad \text{diff. dyn. } \frac{\mathrm{d}}{\mathrm{d}t}\delta\boldsymbol{x} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\delta\boldsymbol{x}. \tag{1}$$

- If there exists positive definite Contraction Metric $\boldsymbol{M}$ such that $\frac{\mathrm{d}}{\mathrm{d}t}V \leq -\alpha V$, the system is contracting.
- $\delta\boldsymbol{x}$ shrinks exponentially in the Riemannian space defined by $\boldsymbol{M}$ (exponential convergence of trajectories).

$$\text{Lya. func. } V = \delta\boldsymbol{x}^\top \boldsymbol{M}(\boldsymbol{x}, t)\delta\boldsymbol{x}, \quad \text{cond. for contraction } \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + 2\,\mathrm{sym}\left(\boldsymbol{M}\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right) \leq -2\alpha\boldsymbol{M} \tag{2}$$

- **Exponential Convergence**: Trajectories globally exponentially converge to each other regardless of initial conditions.
- **Control Design**: Owing to nature of differential dynamics, LPV control design methods can be applied.

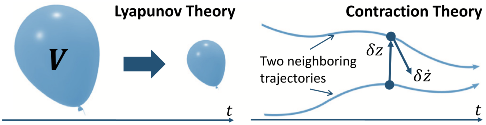

Figure: Lyapunov theory versus contraction theory [2].

Control Contraction Metric (CCM) is an extension of contraction theory to control systems, providing a framework for designing stabilizing controllers [3].

Consider a control-affine system:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t) + \boldsymbol{B}(\boldsymbol{x}, t)\boldsymbol{u} \quad \rightarrow \quad \text{diff. dyn. } \frac{\mathrm{d}}{\mathrm{d}t}\delta\boldsymbol{x} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\delta\boldsymbol{x} + \frac{\partial \boldsymbol{B}}{\partial \boldsymbol{x}}\delta\boldsymbol{x} + \boldsymbol{B}\delta\boldsymbol{u}. \tag{3}$$

Theorem (Control Contraction Metric (CCM) [3])

For a given system, if there exists a bounded metric $\boldsymbol{M}(\boldsymbol{x}, t)$ such that holds

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + 2\,\mathrm{sym}\left(\boldsymbol{M}\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} + \frac{\partial \boldsymbol{B}}{\partial \boldsymbol{x}}\boldsymbol{M}\right) \leq -2\alpha\boldsymbol{M}, \tag{4}$$

for all $\delta\boldsymbol{x} \neq 0$ such that $\delta\boldsymbol{x}^\top\boldsymbol{M}\boldsymbol{B} = 0$, then the system is universally exponentially stabilizable via state feedback.

By using $\boldsymbol{M}$, a feedback controller $\boldsymbol{u}$ can be established considering the closest point on the geodesic between the current state and the target state in the Riemannian space defined by $\boldsymbol{M}$, i.e., $\delta\boldsymbol{u} = k(\boldsymbol{x}, \delta\boldsymbol{x}, u, t)$.

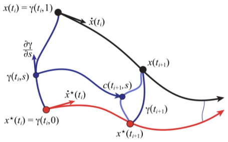

Figure: Control Contraction Metric (CCM) [3].

CCM is similar to the control Lyapunov function (CLF) in that it provides a condition for the existence of a stabilizing controller.

- Both CCM and CLF provide conditions for the existence of stabilizing controllers.
- CCM focuses on the contraction of trajectories in a Riemannian space defined by a metric $\boldsymbol{M}$, while CLF focuses on the decrease of a scalar function V along system trajectories.

There are several limitations as follows:

- Finding a solution of CCM and CLF is non-convex and computationally expensive, (should be solved iteratively and in advance).
- Constraint handling is not straightforward, and it is difficult to guarantee the satisfaction of constraints during the control design process.
- Since CCM and CLF are designed based on global stability properties, they may be conservative.

By defining $\delta\boldsymbol{u} = k \ \rightarrow \ \boldsymbol{u} = K(\boldsymbol{x} - \boldsymbol{x}_d), \ K = \boldsymbol{Y}\boldsymbol{M}$, the contraction metric can be obtained by solving linear matrix inequalities (LMIs) with decision variable $\boldsymbol{W}^{-1} := \boldsymbol{M}$ [4].

According to Theorem 2.4 in [2], by minimizing the condition number χ of $\boldsymbol{M}$, the steady state error can be reduced. The LMI problem can be formulated as follows:

$$\min_{\overline{\boldsymbol{W}},\chi,\nu} \chi + \lambda\nu$$
$$\text{s.t.} \begin{cases} -\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{W}} + 2\,\mathrm{sym}(\mathbb{A}\overline{\boldsymbol{W}}) - 2\nu\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top \preceq -2\alpha\overline{\boldsymbol{W}}, \\ \boldsymbol{I}_n \preceq \overline{\boldsymbol{W}} \preceq \chi\boldsymbol{I}_n, \end{cases} \tag{5}$$

where $\mathbb{A}$ is state-dependent matrix which is employed to transform the system to LPV system, and $\overline{\boldsymbol{W}} = \nu\boldsymbol{W}$ is normalized $\boldsymbol{W}$, and ν is decision variable for magnitude of the contraction metric, and λ is penalty parameter for ν.

The control input can be simply obtained by

$$\boldsymbol{u} = \boldsymbol{u}_d - \boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M}(\boldsymbol{x} - \boldsymbol{x}_d), \quad \text{where } \boldsymbol{M} = \overline{\boldsymbol{W}}^{-1} \cdot \nu \tag{6}$$

- Easy to implement (point-wise convex condition).
- Due to the use of LMIs, it still should be solved in advance.
- According to the objective function, typically, $\boldsymbol{M}$ is designed to be identity matrix and ν is maximized, which results in a simple state feedback controller with a large proportional gain.

Summary of the discussion on CCM and alternative way to use contraction metric $\boldsymbol{M}$:

- CCM is a powerful tool providing the existence of stabilizing controllers.
- CCM has similar problems to CLF, such as non-convexity, computational expense, and difficulty in handling constraints.
- An alternative way to use contraction metric $\boldsymbol{M}$ is to solve LMIs, which is easier to implement but still requires solving in advance and may lead to conservative controllers.

Our approach is . . .

- Solve the contraction problem in an online manner by transforming the contraction condition into a point-wise convex condition in vector space.
- Extend the method to constrained systems, such as input constraints and CBF-motivated constraints.

1 Contraction Theory

2 **Effective Space Approach for Contraction Theory-Based Control Design**
- Main Idea
- Effective Space Approach
- Expected Benefits

3 Conclusion

Main Idea

We first consider how CLF is incorporated into MPC framework.

- The CLF condition is simply added as a constraint in the MPC optimization problem.
- The additional constraints can be easily handled by the optimization solver, and the stability of the closed-loop system can be guaranteed by the CLF condition.
- The stabilizing controller can be obtained point-wise.

To achieve similar properties for contraction theory-based control design, we transform the contraction condition into a point-wise condition.

Using the alterative way, the contraction condition is transformed into point-wise LMI constraints with high computational burden. We project the contraction condition onto the subspace spanned by the error $\boldsymbol{e}$, which results in a point-wise convex condition with low computational burden.

The contraction condition is represented as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + 2\,\mathrm{sym}\left(\boldsymbol{M}\mathbb{A}\right) - 2\boldsymbol{M}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^{\top}\boldsymbol{M} \preceq -2\alpha\boldsymbol{M}. \tag{7}$$

We note that eventually the metric is used as a feedback gain matrix as

$$\boldsymbol{u} = \boldsymbol{u}_d - \boldsymbol{R}^{-1}\boldsymbol{B}^{\top}\boldsymbol{M}\boldsymbol{e}. \tag{8}$$

Therefore, the metric is only used in the subspace spanned by the error $\boldsymbol{e} = \boldsymbol{x} - \boldsymbol{x}_d$.

Thus, we can project the contraction condition onto $\boldsymbol{e}$, which results in a point-wise convex condition with low computational burden.

$$\boldsymbol{y}^{\top}\left(-\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^{\top}\right)\boldsymbol{y} + \boldsymbol{e}^{\top}\left(\frac{1}{T_s} + 2\alpha + 2\mathbb{A}^{\top}\right)\boldsymbol{y} - \frac{\boldsymbol{e}^{\top}\boldsymbol{M}_{\mathrm{pre}}\boldsymbol{e}}{T_s} \leq 0, \tag{9}$$

where $\boldsymbol{y} := \boldsymbol{M}\boldsymbol{e}$ denotes a projected contraction metric onto the $\boldsymbol{e}$, serving as a new optimization variable.

Objective Function (Condition Number): According to Theorem 2.4 in [2], the minimization of the condition number χ of $\boldsymbol{M}$ can be achieved by minimizing the following objective function:

$$J = \|\boldsymbol{y}\|\|\boldsymbol{e}\| - \boldsymbol{y}^\top \boldsymbol{e}. \tag{10}$$

Contraction Condition: Rewrite the projected contraction condition as

$$\boldsymbol{y}^\top \left(-\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\right) \boldsymbol{y} + \boldsymbol{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbb{A}^\top\right) \boldsymbol{y} - \frac{\boldsymbol{e}^\top \boldsymbol{M}_{\mathrm{pre}} \boldsymbol{e}}{T_s} \leq 0. \tag{11}$$

Energy Condition: Minimizing the condition number may result in a very small control input. To avoid this issue, we add the energy condition as follows:

$$\left(\boldsymbol{e}^\top \boldsymbol{y} - \mu \boldsymbol{e}^\top \boldsymbol{e}\right)^2 \leq (\epsilon + s)\|\boldsymbol{e}\|^2, \tag{12}$$

where $\mu \in \mathbb{R}_{>0}$ denotes the desired energy level, and $\epsilon \in \mathbb{R}_{>0}$ and $s \in \mathbb{R}_{\geq 0}$ denote the allowable small energy variation and a slack variable.

Additional Constraints: Any constraints can be added to the optimization problem.

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Resulting Optimization Problem:

$$
\min_{\boldsymbol{y},s} \|\boldsymbol{y}\|\|\boldsymbol{e}\| - \boldsymbol{y}^\top \boldsymbol{e} + \lambda s
$$
$$
\text{s.t.} \begin{cases} \boldsymbol{y}^\top \left(-\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\right)\boldsymbol{y} + \boldsymbol{e}^\top\left(\frac{1}{T_s} + 2\alpha + 2\mathbb{A}^\top\right)\boldsymbol{y} - \frac{\boldsymbol{e}^\top \boldsymbol{M}_{\text{pre}}\boldsymbol{e}}{T_s} \leq 0, \\ \left(\boldsymbol{e}^\top\boldsymbol{y} - \mu\boldsymbol{e}^\top\boldsymbol{e}\right)^2 \leq (\epsilon + s)\|\boldsymbol{e}\|^2, \end{cases} \tag{13}
$$

The tuning parameters are λ and μ.

좀더 생각을..

이산화되면서 하는거라서
그런거야

>> 예제들들들들들
(실험때 생각좀하자)
>> 위성 + 노이즈; 차량(불확실)
메뉴퓰레이터
>> 기존거랑의 차별점이, 장점이 뭐야

>> 실제로 실험으로 어떻게 할지를 먼저
-> 방사형 외란, 초기값

일정은...

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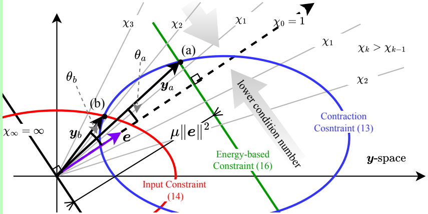
e energy condition.

y level (magnitude of the control input).



n of the proposed method with eclipse-shaped input saturation constraint.

Expected benefits of the proposed method include:

- Contraction metric is not required to be obtained in advance.
- Additional constraints can be easily incorporated into the optimization problem.
- The projected contraction condition is convex and can be solved efficiently.
- The proposed method can be implemented in real-time.

IFAC

- Currently, effective-space formulation and input constraints are considered.
- The input constraint will be removed and the effective-space formulation will be more rigorously justified.

Future work

- Additional constraints, such as state constraints and CBF-motivated constraints, and input constraints will be considered.
- Real-time implementation of the proposed method will be investigated.

Towards Deep Learning

- Using obtained $\boldsymbol{y}$, we can extract the contraction metric $\boldsymbol{M}$.
- Using neural networks, we can learn the mapping from the state $\boldsymbol{x}$ to the contraction metric $\boldsymbol{M}$.
- Hence, if we can justify that the obtained metric is a valid contraction metric under imposed constraints, we can use the learned metric for control design and analysis.

Thank you for your attention!

[1] W. LOHMILLER and J.-J. E. SLOTINE, "On contraction analysis for non-linear systems," *Automatica*, vol. 34, no. 6, pp. 683–696, 1998. [Online]. Available: https://doi.org/10.1016/S0005-1098(98)00019-3

[2] H. Tsukamoto, S.-J. Chung, and J.-J. E. Slotine, "Contraction theory for nonlinear stability analysis and learning-based control: A tutorial overview," *Annual Reviews in Control*, vol. 52, pp. 135–169, 2021. [Online]. Available: https://doi.org/10.1016/j.arcontrol.2021.10.001

[3] I. R. Manchester and J.-J. E. Slotine, "Control contraction metrics: Convex and intrinsic criteria for nonlinear feedback design," *IEEE Transactions on Automatic Control*, vol. 62, no. 6, pp. 3046–3053, 2017. [Online]. Available: https://doi.org/10.1109/TAC.2017.2668380

[4] H. Tsukamoto, S.-J. Chung, and J.-J. E. Slotine, "Neural stochastic contraction metrics for learning-based control and estimation," *IEEE Control Systems Letters*, vol. 5, no. 5, pp. 1825–1830, 2021. [Online]. Available: https://doi.org/10.1109/LCSYS.2020.3046529